Chemosphere 271 (2021) 129591

Contents lists available at ScienceDirect

Chemosphere

journal homepage: www.elsevier.com/locate/chemosphere

Size-dependent chronic toxicity of fragmented polyethylene microplastics to *Daphnia magna*

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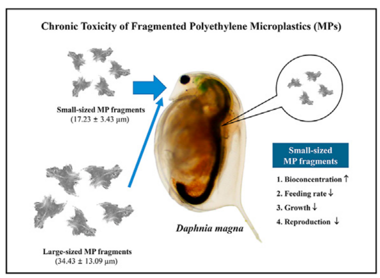
HIGHLIGHTS

- Accumulation of microplastic (MP) fragments was higher than that of MP beads.
- This likely resulted in higher mortality in *Daphnia magna* for MP fragments.
- MP fragments decreased the carbohydrate and protein reserves of *D. magna.*
- Inhibition of feeding in *D. magna* depended on the size of MP fragments.
- This may lead to size-dependent reduction in growth and reproduction of *D. magna.*

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Received 30 October 2020
Received in revised form
29 December 2020
Accepted 5 January 2021
Available online 8 January 2021

Handling Editor: Michael Bank

Keywords:
Daphnid
Fragmentation
Feeding rate
Microplastic
Mortality
Reproduction

ABSTRACT

Fragmented microplastics (MPs) are emerging contaminants in freshwater environments; however, long-term assessment of their toxicity remains limited. This study aimed to evaluate and compare the chronic toxicity (21 d) of synthesized polyethylene MP fragments and commercial polyethylene MP beads to *Daphnia magna*. Ingestion of small- and large-sized MP fragments (17.23 ± 3.43 and 34.43 ± 13.09 μm, respectively) by *D. magna* was significantly ($p < 0.05$) higher, by 8.3 and 5.2 times, respectively, than that of MP beads (39.54 ± 9.74 μm). The survival of *D. magna* exposed to small- and large-sized MP fragments (20 and 60%, respectively) was significantly ($p < 0.05$) lower than that of individuals exposed to MP beads (90%). In particular, small-sized MP fragments significantly ($p < 0.05$) reduced algal feeding (from 95% to 76%), body length (from 4.20 mm to 3.98 mm), and the number of offspring (from 109 to 74) in *D. magna*, when compared with MP beads, likely due to their longer retention time and greater interference in the digestive tract. These findings suggest that fragmentation of MPs into μm-scale particles can pose a significant ecological risk to aquatic organisms; moreover, further studies are required to identify the underlying toxicity mechanism.

1. Introduction

Plastics are widely produced for industrial and domestic applications because of their unique properties (Wang et al., 2016). Plastic production has increased globally over the last few decades, reaching 359 million tons in 2018 (PlasticsEurope, 2019).

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https://doi.org/10.1016/j.chemosphere.2021.129591
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Microplastics (MPs) are defined as plastic particles less than 5 mm in diameter (Thompson et al., 2009; Eerkes-Medrano et al., 2015; Lambert and Wagner, 2018). Secondary MPs, including fibers and fragments, can be produced by photolytic and biological degradation and oxidative weathering processes, leading to their widespread presence in aquatic systems (Browne et al., 2007; Eerkes-Medrano et al., 2015). In particular, Lasee et al. (2017) reported that MP (53–105 μm) concentrations in contaminated lakes and wetlands of Texas, USA were as high as 1.56 ± 1.64 and 5.51 ± 9.09 mg L^{-1}, respectively. Fibers and fragments account for approximately 79% of the MPs in global freshwaters (Li et al., 2020). Hidayaturrahman and Lee (2019) reported that fragmented MPs account for 31.4–53.4% of the total MPs in the influent of wastewater treatment plants in Daegu, Republic of Korea. In addition, several studies have reported that more than 90% of the MPs in freshwater effluents and surface waters measure less than 20 μm (Ivleva et al., 2017; Schymanski et al., 2018; Triebskorn et al., 2019).

Numerous studies have assessed the adverse effects of MPs in freshwater ecosystems (Browne et al., 2007; Eriksen et al., 2013; Eerkes-Medrano et al., 2015; Du et al., 2020; Xu et al., 2020). However, most of these studies focused on acute toxicity (Ma et al., 2016; Rehse et al., 2016; Rist et al., 2017; Ziajahromi et al., 2017; Na et al., 2021), and chronic toxicity studies have been limited to uniformly shaped commercial primary MPs (e.g., microbeads) (Besseling et al., 2014; Rist et al., 2017; Pacheco et al., 2018; Bosker et al., 2019; Eltemsah and Bøhn, 2019), which is not a realistic scenario in ambient waters (Li et al., 2020). Few studies have reported the chronic toxicity of fragmented secondary MPs in *Daphnia magna* (Ogonowski et al., 2016; Jaikumar et al., 2019) in comparison with spherical primary MPs. Ogonowski et al. (2016) reported that secondary polyethylene (PE) MPs decreased reproduction in *D. magna* when compared with primary spherical fluorescent MPs (1–5 μm; Cospheric LLC, Goleta, USA) upon exposure for 21 days. In contrast, Jaikumar et al. (2019) documented higher chronic toxicity potential of the MP beads than PE MP fragments in cladocerans. Moreover, the primary MPs used in both these studies had higher density (1.3 g cm^{-3}) than the secondary PE MPs (<1 g cm^{-3}) used for toxicity comparison. These findings highlight the necessity of a comparative chronic toxicity study using the same type of primary and secondary MPs.

MPs occur in various shapes (e.g., pellets, fibers, and fragments), leading to varying toxicity potential towards aquatic organisms (Ogonowski et al., 2016; Gray and Weinstein, 2017; Ziajahromi et al., 2017; Qiao et al., 2019; Na et al., 2021). For example, Na et al. (2021) reported that the acute toxicity (48 h) of fragmented PE MPs (37.24 ± 11.76 μm) was 80 times higher than that of PE MP beads (37.05 ± 3.96 μm), possibly due to the irregular shape and high specific surface area of the former. Unlike microbeads, weathered MPs can cause damage and morphological changes in the digestive tract of aquatic organisms (Gray and Weinstein, 2017; Qiao et al., 2019). Moreover, particle size plays an important role in determining MP ingestion and toxicity (Andrady, 2011; Wagner et al., 2014). For example, Cole et al. (2013) reported that smaller polystyrene (PS) microbeads (7.3 μm) significantly reduced algal feeding (24 h) in the marine copepod *Centropages typicus* when compared with larger PS microbeads (20.6 μm).

Therefore, the aim of the present study was to assess and compare the chronic toxicity of PE MP fragments and beads in *D. magna*. Two different sizes of MP fragments were prepared using PE pellets to evaluate the effect of MP size on *D. magna* feeding behavior, survival, growth, and reproduction. The filter-feeding crustacean *D. magna* was used as a model species because it plays an important role in freshwater ecosystems and is widely used in ecotoxicological studies (Watanabe et al., 2007; Liu et al., 2019; Bownik et al., 2019; Xu et al., 2020). We hypothesized that smaller fragmented MPs would have higher chronic toxicity in *D. magna* because of greater disruption of ingestion and digestion, leading to reduced survival, growth, and reproduction.

2. Materials and methods

2.1. Preparation and characterization of MPs

MP fragments were prepared from PE pellets (density = 0.952 g cm^{-3}, Sigma Aldrich, USA) using a laboratory mixing extruder (Dynisco, USA). The PE pellets were melt-blended at a constant barrel and die temperature of 170 °C and rotor speed of 10 rpm to obtain plastic fiber extrusions. The plastic fibers were then cryogenically ground in liquid nitrogen in Freezer/Mill 6875 (SPEX® SamplePrep, US) to obtain MP fragments. These MP fragments were sieved through a steel mesh of 25 and 53 μm mesh size to yield small- and large-sized MP fragments, respectively. PE MP beads (density = 0.94 g mL^{-1}, 40–48 μm) were purchased from Sigma Aldrich (USA) for comparison.

The size and shape of the MPs were analyzed by field emission-scanning electron microscopy (FE-SEM; Quanta™ 250 FEG, FEI, USA). The chemical composition of MPs was also determined by Fourier transform-infrared (FT-IR) spectroscopy with attenuated total reflection (Cary 630, Agilent, USA) to identify any contamination.

2.2. Chronic toxicity test

Daphnia magna, cultured at the Water Quality Research Laboratory of Korea University (Republic of Korea) since 2010, were obtained from the National Institute of Environmental Research (Republic of Korea). Daphnids were grown in a 2-L beaker containing M4 medium (pH = 7.8 ± 0.1, hardness = 250 ± 25 mg L^{-1} $CaCO_3$) at 20 ± 2 °C under a 16:8 h light:dark photoperiod, following the guidelines of Test No. 211 of the Organization for Economic Co-operation and Development (OECD; 2012). Animals were fed daily with *Chlorella vulgaris* (5×10^5 cells mL^{-1}), and the M4 medium was renewed twice per week. The neonates (<24-h-old) were isolated and cultured for 3 days to obtain juvenile daphnids for chronic toxicity tests.

The MP stock solution (10 mg L^{-1}) was prepared in M4 medium with Tween 80 (0.001% v/v; Sigma Aldrich, USA) as a surfactant to prevent MP aggregation. The number of particles per liter was determined using a hemocytometer under a microscope (Olympus CX33, Tokyo, Japan) and was noted as $3.4 \pm 1.8 \times 10^6$, $1.7 \pm 1.1 \times 10^7$ and $2.1 \pm 1.0 \times 10^7$ (n = 15) particles L^{-1} for MP beads, large- and small-sized MP fragments, respectively.

Chronic toxicity tests (n = 10) were performed following the OECD Test No. 211 (OECD, 2012), with one juvenile *D. magna* (4-day-old) in a glass beaker containing a 50-mL volume of M4 medium with 0.001% (v/v) Tween 80 and 5 mg L^{-1} MPs. Tests were conducted at 20 ± 2 °C under a 16:8 h light:dark photoperiod and the medium was replaced twice per week. Daphnids were fed daily with *C. vulgaris* at a concentration of 5×10^5 cells mL^{-1}. The M4 medium with 0.001% (v/v) Tween 80 was used as a control. The chronic toxicity test continued for 21 days, during which the survival, body length of adults and offspring, time to first brood, brood size and number, and total number of offspring per female were evaluated. The tests were valid only when mortality was not observed in the control and the total number of offspring per female was larger than 60 (OECD, 2012).

2.3. Feeding rate analysis

The MP exposure conditions for feeding analysis (n = 3) were the same as those for chronic toxicity testing, except for exposure duration of 48 h. Before and after exposure, 10-μL sample aliquots were collected after stirring the medium for 30 s. The number of algal cells in the medium was determined using a hemocytometer under a microscope (Olympus CX33, Tokyo, Japan), and the feeding rate (F) was calculated using Eq. 1

$$F\ (\%) = \frac{(N_0 - N_t)}{N_0} \times 100 \tag{1}$$

where N_0 and N_t represent the number of *C. vulgaris* cells mL^{-1} at 0 and 48 h, respectively.

During the feeding rate test (48 h), *D. magna* were photographed using a microscope (Olympus C×33, Tokyo, Japan) equipped with a digital camera (3.1 M P 1/2.8″ Sony Exmor CMOS Sensor).

2.4. Bioconcentration of MPs

The MP exposure conditions for bioconcentration analysis (n = 3) were the same as those for the chronic toxicity test, except for the exposure duration (6 days) and number of daphnids (16 in 800 mL of medium). After exposure, daphnids were rinsed with clean medium to remove MPs from their carapace and the adherent medium was gently removed using Kimtech tissue paper. Animals were weighed using an ultra-micro balance (RADWAG AS 220.R2, Poland) and then frozen immediately at −80 °C.

Following Na et al. (2021), 10 frozen daphnids were digested in 15 mL KOH (10%) for 24 h at 60 °C in triplicate. The digested samples

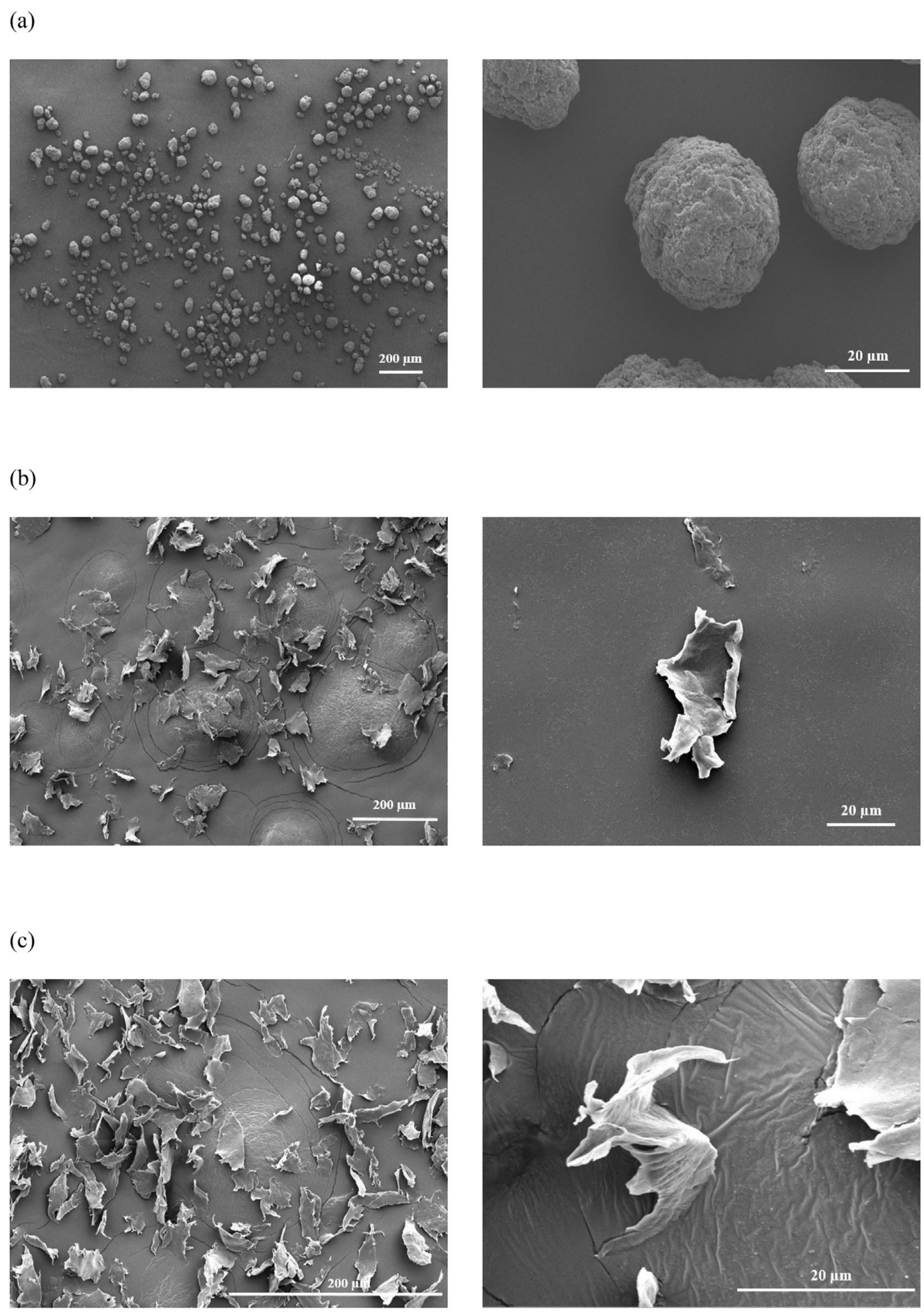

Fig. 1. Scanning electron micrographs of microplastic (MP) (a) beads (39.54 ± 9.74 μm) and (b) large- (34.43 ± 13.09 μm) and (c) small-sized (17.23 ± 3.43 μm) MP fragments.

were homogenized, and 5 μL of the homogenate was analyzed under a microscope (Olympus CX33, Tokyo, Japan) equipped with a digital camera (3.1 M P 1/2.8″ Sony Exmor CMOS Sensor), with five technical replicates.

2.5. Energy reserve analysis

The MP exposure conditions for the energy reserve analysis (n = 3) were the same as those for bioconcentration testing. Ten frozen daphnids were homogenized in phosphate buffered saline (PBS; pH = 7.4; 0.137 M NaCl, 0.002 M KCl, 0.010 M Na_2HPO_4, 0.002 M KH_2PO_4) for carbohydrate and protein analyses. The energy reserves of *D. magna* were calculated using energetic equivalents (17.5 kJ per g carbohydrate, 24 kJ per g protein), following Gnaiger (1983).

The carbohydrate content was determined following the procedure described by De Coen and Janssen (1997). Briefly, the homogenate (250 μL) was mixed with 30 μL of 15% trichloroacetic acid (TCA) and incubated at −20 °C for 10 min. The mixture was centrifuged at 3,000×*g* for 10 min at 4 °C and the supernatant was collected. The residual pellet was washed with 30 μL of 5% TCA and both supernatant fractions were combined. The supernatant was mixed with 5% phenol and 98% H_2SO_4 at a volume ratio of 1:2:4 and incubated at 20 °C for 30 min. Absorbance was measured at 492 nm, and the carbohydrate content was calculated using a glucose standard curve.

The protein content was determined using Bradford's reagent (Bradford, 1976). Briefly, the homogenate (250 μL) was centrifuged at 16,000×*g* for 15 min at 4 °C. The supernatant (25 μL) was collected and mixed with 75 μL of PBS and 2.5 mL Bradford's reagent (0.01% Coomassie Brilliant Blue G-250, 4.7% ethanol, and 8.5% phosphoric acid; w/v). The sample was allowed to stand for 15 min after mixing and absorbance was measured at 595 nm. The protein content was determined using a bovine albumin standard curve.

2.6. Statistical analyses

All statistical analyses were performed using SAS version 9.4 (SAS Institute Inc., Cary, USA). All treatment groups were analyzed using one-way analysis of variance (ANOVA) and post-hoc Tukey's honest significance test. Results were considered significant for $p < 0.05$.

3. Results and discussion

3.1. Ingestion of MP fragments

The scanning electron micrographs of the MP beads and fragments are shown in Fig. 1. MP beads occurred as regular spheres measuring 39.54 ± 9.74 μm (n = 20). Both large- and small-sized MP fragments occurred in irregular rough shapes measuring 34.43 ± 13.09 and 17.23 ± 3.43 μm (n = 20), respectively. The FT-IR spectra of the MP beads and fragments indicated that they were all composed of polyethylene (Fig. S1), exhibiting three specific bands (Hummel, 2012), i.e., band 1 at 725 cm^{-1} (CH_2 rocking), band 2 at 1460 cm^{-1} (CH_2 stretching), and band 3 around 2865 and 2920 cm^{-1} (CH stretching).

The ingestion of MP beads and fragments by *D. magna* for 6 days is shown in Fig. 2. The concentration of MP fragments in the daphnids was significantly ($p < 0.05$) higher than that of MP beads, with large- and small-sized MP fragments exhibiting 5.2 and 8.3 times higher concentrations than the beads, respectively. In general, daphnids non-selectively ingest particles measuring less than 90 μm (Rehse et al., 2016; Scherer et al., 2017), suggesting that the higher bioconcentration may have been caused by reduced elimination. The larger aspect ratio of MP fragments, when compared with that of MP beads, may allow easy accumulation of these particles in tissues, leading to higher bioconcentration (Hurley et al., 2017). In addition, smaller particles can cross cellular barriers more easily than larger particles, likely resulting in higher bioconcentration (Kato et al., 2003; Oberdörster et al., 2005). Frydkjær et al. (2017) reported that < 1% of *D. magna* could eliminate MP fragments (10−75 μm) from their gut, whereas 49% of *D. magna* could clear MP beads (10−106 μm) within 24 h. Gray and Weinstein (2017) reported that grass shrimp (*Palaemonetes pugio*) accumulated MP fragments (63.5 μm) at greater concentrations than MP beads (80.4 μm), possibly because of the higher residence time of the MP fragments in the gut. Qiao et al. (2019) also reported that PS fragments (4−40 μm) were more concentrated than PS beads (15 μm) in zebrafish gut, likely because of the longer gut passage time of PS fragments. These findings suggest that greater bioconcentration of small-sized MP fragments may lead to higher chronic toxicity than large-sized MP fragments and MP beads.

3.2. Chronic toxicity of MP fragments

The survival of *D. magna* exposed to MP beads and fragments for

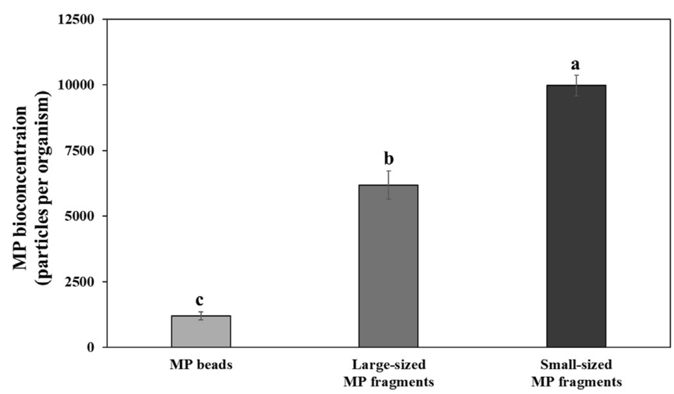

Fig. 2. Bioconcentration of microplastic (MP) beads (39.54 ± 9.74 μm) and large- (34.43 ± 13.09 μm) and small-sized (17.23 ± 3.43 μm) MP fragments in *Daphnia magna* for 6 days. The MP concentration was 5 mg L^{-1}. Data are presented as mean ± standard deviation (n = 3). Different lowercase letters above the bars indicate significant differences ($p < 0.05$).

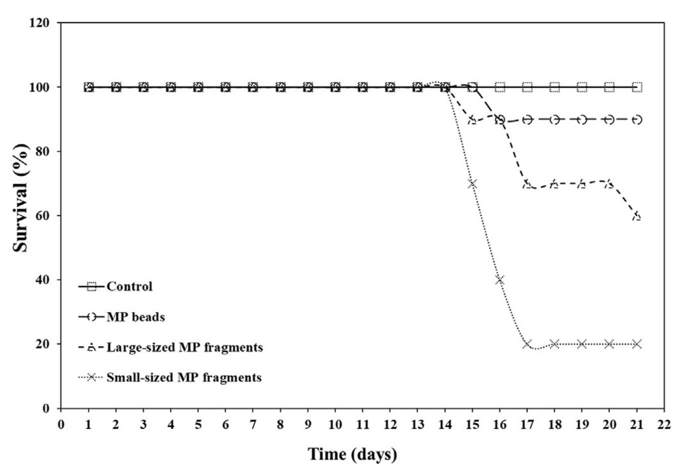

Fig. 3. Survival of *Daphnia magna* exposed to microplastic (MP) beads (39.54 ± 9.74 μm) and large- (34.43 ± 13.09 μm) and small-sized (17.23 ± 3.43 μm) MP fragments for 21 days. The MP concentration was 5 mg L^{-1}. The control treatment consisted of Tween 80 in M4 medium (0.001%, v/v).

21 days is illustrated in Fig. 3. Exposure to MP fragments resulted in significantly ($p < 0.05$) lower survival than exposure to MP beads (90%), with small-sized MP fragments (20%) resulting in considerably lower survival than large-sized MP fragments (60%). Ogonowski et al. (2016) reported a 50% reduction in *D. magna* survival on exposure to PE fragments (2.6 ± 1.8 μm) for 21 days when compared with exposure to MP beads (4.1 ± 1.0 μm), which could be attributed to the two-fold greater retention time of PE fragments in the gut. Unlike the control and MP beads, MP fragments blocked the gut in dead *D. magna*, and substantial contraction of the digestive system was observed for small-sized MP fragments (Fig. S2). Jemec et al. (2016) also observed gut blockage by polyethylene terephthalate fibers (62–1,400 μm) in all dead daphnids. These findings suggest that accumulation of MP fragments in the gut may lead to mortality in *D. magna*.

The growth and reproduction of *D. magna* exposed to MP beads and fragments for 21 days are illustrated in Fig. 4. Only small-sized MP fragments led to a significant ($p < 0.05$) reduction in the body length and total number of offspring per female (see Fig S3 for clarity) of exposed *D. magna* when compared with the control. This reduction in fecundity could be attributed to the reduced number of broods, as there was no significant ($p > 0.05$) difference in the brood size and time to first brood (Fig. S4). Moreover, small-sized MP fragments significantly ($p < 0.05$) reduced the body length of offspring in the fourth and fifth broods (Fig. S5). Ogonowski et al. (2016) reported lower reproduction in *D. magna* exposed to PE fragments (2.6 ± 1.8 μm) for 21 days when compared with those exposed to MP beads (4.1 ± 1.0 μm). Ziajahromi et al. (2017) further reported that exposure to PE fibers (26–1,150 μm) at 500 μg L^{-1} for 8 days significantly reduced adult body length and total number of offspring in *Ceriodaphnia dubia*, whereas a similar effect was observed at 1,000 and 2,000 μg L^{-1} for PE beads (1–4 μm). These findings suggest that MP fragmentation may reduce growth and reproduction in *D. magna*, likely by feeding inhibition (Au et al., 2015).

The inhibition of feeding in *D. magna* exposed to MP beads and fragments for 48 h is illustrated in Fig. 5. Only small-sized MP fragments significantly ($p < 0.05$) reduced feeding by 76% when compared with the control (94%). Nasser and Lynch (2016) demonstrated that inhibition of feeding in *D. magna* exposed to COOH-PS nanoparticles (293.8 nm) might be caused by residual particles in the gut. Accumulation of MP fragments in the gut of *D. magna* was observed on exposure for 48 h, whereas the gut of control individuals and those exposed to MP beads was empty (Fig. S6). Furthermore, the accumulation of small-sized MP fragments in the gut occurred earlier (at 24 h) than that of large-sized MP fragments (at 48 h). Feeding inhibition due to MP fragments affects the metabolism of aquatic organisms and may lead to energy depletion (Welden and Cowie, 2016; Murphy and Quinn, 2018). The carbohydrate and protein energy reserves of *D. magna* exposed to MP beads and fragments for 6 days are illustrated in Fig. 6. Both small- and large-sized MP fragments led to a considerable decrease in these reserves when compared with the control treatment and MP beads. As energy reserves are essential for somatic maintenance, growth, and reproduction of organisms (Sprung, 1991), the observed reduction in growth and reproduction of *D. magna* could be attributed to the depletion of energy reserves.

Although MPs are ubiquitous in freshwater ecosystems, the long-term impact of MPs on freshwater organisms remains poorly understood. Environmental MPs are fragmented into smaller

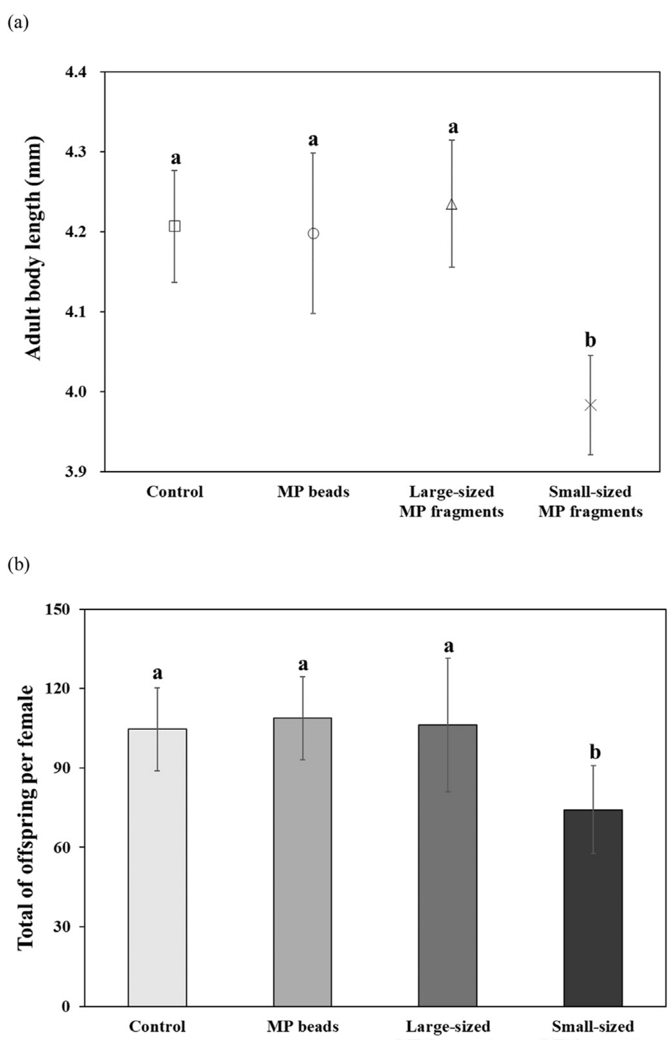

Fig. 4. (a) Growth and (b) reproduction of *Daphnia magna* exposed to microplastic (MP) beads (39.54 ± 9.74 μm) and large- (34.43 ± 13.09 μm) and small-sized (17.23 ± 3.43 μm) MP fragments for 21 days. The MP concentration was 5 mg L^{-1}. The control treatment consisted of Tween 80 in M4 medium (0.001%, v/v). Data are presented as mean ± standard deviation (n = 10). Different lowercase letters above the bars indicate significant differences ($p < 0.05$).

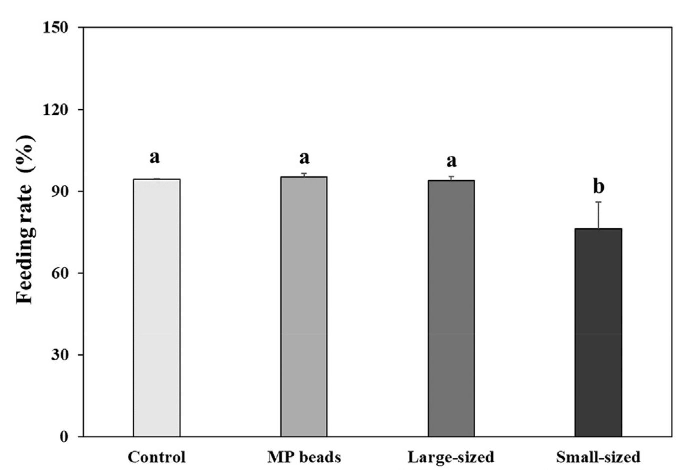

Fig. 5. Feeding rate of *Daphnia magna* exposed to microplastic (MP) beads (39.54 ± 9.74 μm) and large- (34.43 ± 13.09 μm) and small-sized (17.23 ± 3.43 μm) MP fragments for 48 h. The MP concentration was 5 mg L^{-1}. The control treatments consisted of Tween 80 in M4 medium (0.001%, v/v). Data are presented as mean ± standard deviation (n = 3). Different lowercase letters above the bars indicate significant differences ($p < 0.05$).

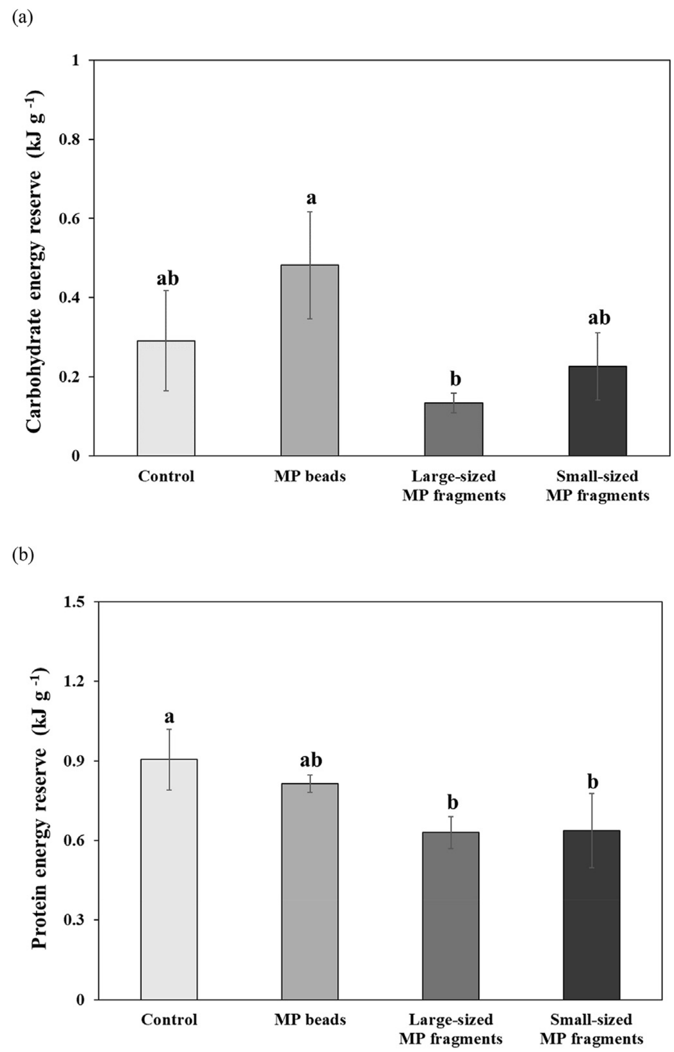

Fig. 6. (a) Carbohydrate and (b) protein energy reserves of *Daphnia magna* exposed to microplastic (MP) beads (39.54 ± 9.74 μm) and large- (34.43 ± 13.09 μm) and small-sized (17.23 ± 3.43 μm) MP fragments for 6 days. The MP concentration was 5 mg L^{-1}. The control treatment consisted of Tween 80 in M4 medium (0.001%, v/v). Data are presented as mean ± standard deviation (n = 3). Different lowercase letters above the bars indicate significant differences ($p < 0.05$).

particles by physical and chemical weathering processes (Browne et al., 2007; Eerkes-Medrano et al., 2015; Li et al., 2020). This study suggests potential chronic toxicity of MP fragments having the size less than small MPs (25–1000 μm; Eo et al., 2018). The adverse effects of MPs may include changes at morphological and molecular levels in addition to life history (Imhof et al., 2017), may leading to reduced fitness in *D. magna*. Given that *D. magna* plays an important role as food sources for higher trophic level aquatic organisms (Shaw et al., 2008), this impact may transfer further in freshwater ecosystems.

4. Conclusions

MP fragments exhibited higher chronic toxicity to *D. magna* than MP beads, which could be attributed to the greater disruption of algal ingestion and digestion by the former, leading to reduced survival, growth, and reproduction. This adverse effect was greater for small-sized MP fragments (17.23 ± 3.43 μm) with higher bioconcentration potential in *D. magna*, compared with large-sized MP fragments (34.43 ± 13.09 μm). Although the underlying toxicity mechanism of MP fragments, including intestinal damage in *D. magna*, could not be clearly identified, this study demonstrates the long-term ecological risk of MP fragmentation in aquatic environments.

Credit author statement

Dahee An: Data curation, Formal analysis, Investigation, Writing - original draft. **Joorim Na**: Methodology, Investigation. **Jinyoung Song**: Methodology, Investigation. **Jinho Jung**: Conceptualization, Funding acquisition, Supervision, Writing - Review & Editing

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was supported by a grant from the National Research Foundation of Korea (NRF), funded by the Korean Government (NRF-2019R1A2C1002890).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2021.129591.

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